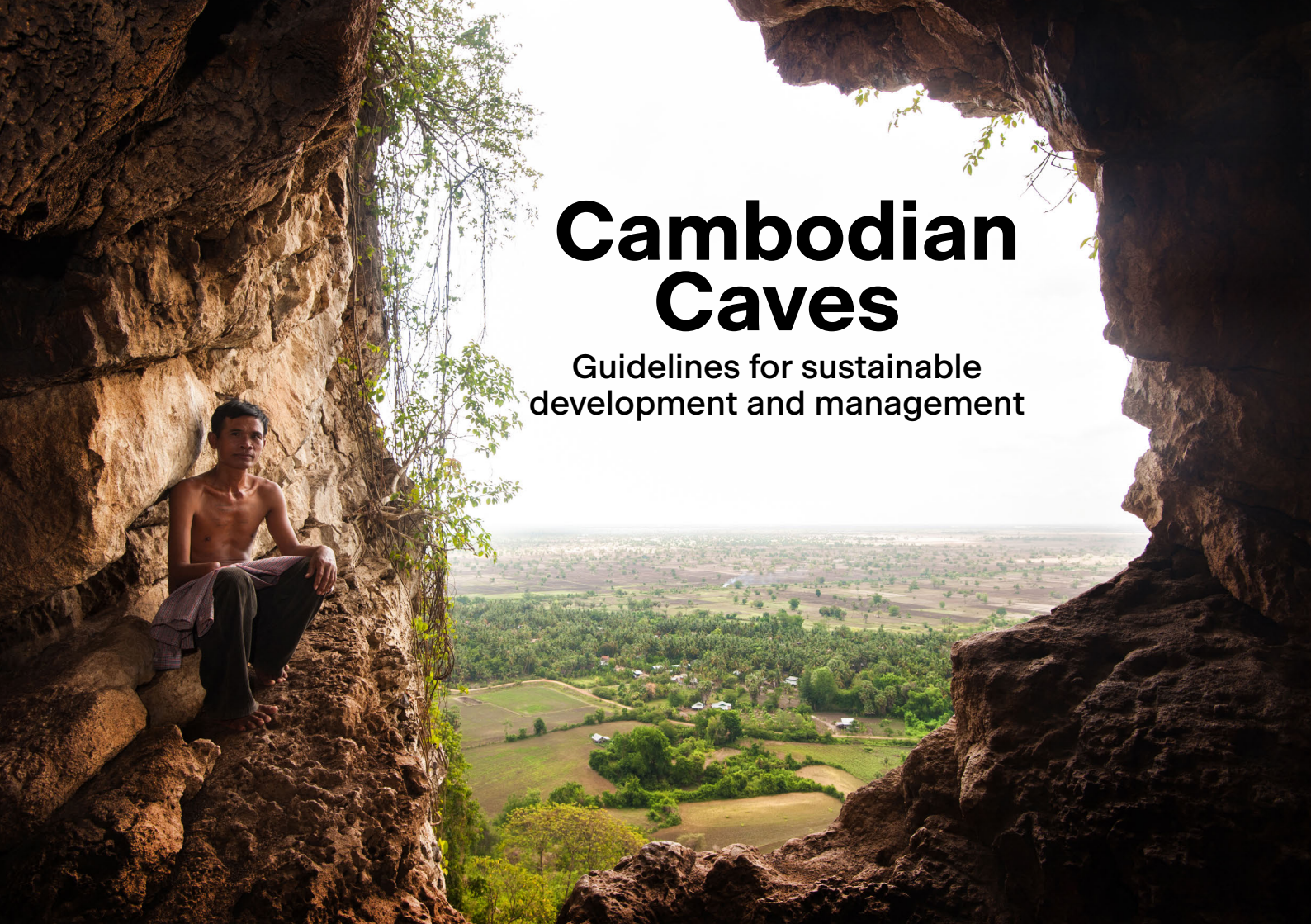


Cambodian Caves

Guidelines for sustainable development and management



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Funded by the European Union

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CAMBODIAN CAVES

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Preface

Cambodia's limestone caves are among the country's most under-appreciated natural treasures. Hidden within dramatic karst landscapes, they hold exceptional cultural, biological, and economic value. These ecosystems support unique and often irreplaceable lifeforms, preserve remnants of our past, and continue to sustain local livelihoods through tourism and guano collection. Yet despite their significance, cave environments remain poorly understood, largely unregulated, and increasingly threatened.

These guidelines were developed in response to a growing need for practical, accessible tools to inform the sustainable development and conservation of Cambodian caves. They are intended to support local authorities, protected area managers, and other decision-makers in balancing the use and protection of caves - ensuring that communities can benefit from them today without compromising their future. They draw on global best practice and are based on the realities of the situation in Cambodia. There is now an urgent need to safeguard bat populations, plus the unique biodiversity, cultural heritage, and vital ecosystem services that caves provide. These guides are not exhaustive; we are still learning about the dynamics of cave systems; how best to conserve them. But they provide a clear starting point for better decision-making, invite further engagement and collaboration, and give resources for those seeking to learn more.

Protecting Cambodia's caves is not just about conserving rare species or ancient artefacts, but about recognising their place in the country's natural and cultural fabric, and ensuring that these remarkable places can continue to inspire, educate and support generations to come.

Pablo Sinovas - Fauna & Flora



Karst landscape at La Ang Pracheav, Phnom Romseisak, Battambang Province © Chourn Phyroum Fauna & Flora

Context

Limestone karst regions and their caves are internationally recognised for their deep cultural significance and extraordinary biodiversity, many aspects of which have enormous economic value[1-3]. Because of this, karst ecosystems form an irreplaceable component of protected area networks in many Southeast Asian countries, including World Heritage Sites, Global Centres of Plant Diversity and numerous sites of global biodiversity importance. Until recently however, no limestone karst area in Cambodia was included in a nationally protected area or otherwise legally protected for its natural values[4]. This omission is mirrored in the fact that limestone karst ecosystems are only mentioned once in the 2016 National Biodiversity Strategy and Action Plan for Cambodia[5].

Due to their solubility, limestone areas support the vast majority of caves in Cambodia. While the total extent of limestone rocks in-country is not accurately known[6], existing data indicate that over 95 karst outcrops comprising ≈120 km^2 occur in the kingdom[7]. The largest areas of karst occur in the western provinces of Battambang and Banteay Meanchey (≥ 45 hills) where karst hills occur intermittently along a ≈250 km belt or archipelago that extends into Thailand. Many isolated karst outcrops also occur in the southern provinces of Kampot and Kep (≥ 40 hills) which extend into Vietnam, and a lesser number in the northern Stung Treng Province (≥ 10 hills), close to the Laos border[6,7]. Over 320 caves have been registered in these and other smaller areas of karst to date[8], although the true number is considerably greater.



Given their cultural importance and remarkable biodiversity, there is a desperate need for sustainable cave management practices in Cambodia. This is elevated by their inherent vulnerability and the many threats they face[4]. As such, the purpose of these guidelines is to provide local authorities with introductory information on the importance of caves, cave ecosystems and life, typical conservation threats and mitigation, and finally, important practices for effective cave management. Since this work does not include aboveground karst ecosystems and can only briefly touch on some of the many fields and disciplines involved in understanding and sustainably managing caves, further sources of online information and references are provided for those interested in learning more.



Sixth-century Indian temple in Vihear Tuk-Bonn, Phnom Chhngauk, Kampot Province © Neil Furey Fauna & Flora



Cultural, Biodiversity & Economic Importance of Caves

Karst and caves retain important pieces of the Earth's history, including its geological evolution, palaeo-environments and past climates, extinct megafauna and the origin and rise of human civilisations. Caves in Southeast Asia and Cambodia have been used by humans for many purposes for countless millennia and often provide the only evidence regarding ancestral occupation of modern territories. This is because caves provide ideal conditions for preserving tools, food remains and other material remnants of ancient communities, whereas those left aboveground tend to decay rapidly in the humid tropics. For example, one of the first well-excavated prehistoric sites in Cambodia, Laang Spean cave in Battambang, contains human artefacts dating back to over 71,000 years before present and provides important insights to human prehistory in Southeast Asia, including our evolution from living as hunter-gatherers to a settled agricultural way of life[9,10].

A more recent example is the presence of late 6th century temples in caves in Kampot thought to be built by Indian settlers that may have arrived by sea. Unlike the Angkorian-era (ca. 9th to 15th century) temples and artefacts in Siem Reap, where Hindu and Buddhist influences are blended with Cambodian culture, these appear to be solely based on Indian art and provide important clues to the origins of Khmer culture[11]. More broadly, the continued importance of caves to Cambodian culture is reflected in their nationwide use as Buddhist shrines. This association is fundamental and dates back to the beginning

◁ Isolated limestone karst hill in Kampot Province, Cambodia © Nehru Pry

of the Buddhist tradition over 2,500 years ago[12], such that countless temples and shrines are located in limestone karst caves throughout Southeast Asia and Cambodia today. Many of these sites have since become famous tourism attractions in their own right, such as Phnom Sampeu in Battambang and Phnom Chhngauk in Kampot.

The importance of Southeast Asian karst regions as 'biodiversity arks' or refuges that possess high species diversity and levels of endemism is now widely recognised[1–3]. While karst is still poorly studied in Cambodia, recent research has found many previously undescribed reptile and snail species and strongly suggests that additional taxa await discovery in the many isolated outcrops and caves that have yet to be explored[13–16]. The existing data also suggest that many of these species have very limited distributions, such that conservation of individual outcrops will be essential for their survival. The same is true for underground life in karst, which mostly consists of invertebrates. Unlike most surface-dwelling species, many of the invertebrates that permanently live in caves are highly range-restricted endemics incapable of dispersing to other sites. Because these are highly susceptible to environmental change and consequently extinction, caves rank among the hottest of biodiversity hotspots worldwide in terms of their levels of species endemism and threat[17].

Caves and bats are often associated in the minds of many people. Rightly so, since caves and other underground sites often shelter significant portions of national bat faunas[18] and roughly half of the 80+ bat species now known in Cambodia[4,19]. These include many bat species that provide economically-significant services by controlling serious agricultural pests, pollinating major crop species, dispersing seeds and producing valuable plant fertilizer (guano), particularly at local levels. For instance, the cave nectar bat (*Eonycteris spelaea*) is now recognised as the primary pollinator of durian and several other important plant species[20–22]. Dietary studies have shown that cave nectar bats in Cambodia visit many of the same plants as in Thailand and Peninsular Malaysia and thereby contribute to maintaining durian farming and coastal mangroves within the kingdom[23].



▷ *Clostophis udayaditinus*, a new species recently discovered in karst areas of Battambang Province © Somsak Panha Fauna & Flora



CAMBODIAN CAVES

As another example, seven enormous cave colonies representing ≈6.4 million wrinkle-lipped bats (*Mops plicatus*) are currently known in Cambodia, five of which are in Battambang and two in Kampot. Protection of these colonies by local communities has allowed them to harvest huge quantities of potent plant fertilizer for decades. And should conservative estimates of the nightly consumption of just one of the many pest species eaten by these bats in Thailand hold true for Cambodia, the seven cave colonies prevent the loss of over 2,300 tons of rice annually. This is equivalent to daily rice meals for over 20,800 Khmer each year[24].

The evening emergence of these colonies is also spectacular, such that one site in Battambang has consistently ranked as the top attraction for tourists for many years. As a consequence, these cave bat colonies directly contribute to supporting the livelihoods of the many thousands of people engaged in the tourism and hospitality sectors within the Battambang Province.

◁ *Cyrtodactylus laangensis*, a new gecko species from caves in Phnom Laang, Kampot Province © Jeremy Holden



Cyrtodactylus kampingpoiensis, a new species discovered in Kamping Poi karst hill, Battambang © Fauna & Flora



Pseudoscorpion sp.
© Ana Komericki Fauna & Flora



Wrinkle-lipped bats (*Mops plicatus*) form the largest bat colonies in karst caves in Cambodia © MerlinTuttle.org



Cave nectar bat (*Eonycteris spelaea*) pollinating a durian flower © MerlinTuttle.org

Cave Environments & Life

While highly durable, limestone rocks are susceptible to chemical erosion by even mildly acidic waters in the form of rainfall and percolating waters. These create countless passages which progressively enlarge along lines of weakness inside the rock, while sculpting the exterior into the dramatic karst formations for which Southeast Asia is famous. As a result, limestone karst areas are typically full of subterranean caverns and passageways and these have supported the evolution of life unlike any found in the surface world. Being highly porous, karst areas also often lack an aboveground pattern of drainage and although their surfaces quickly dry after rainfall, the water below can circulate for a long time through voids inside the rock.

Caves are usually the best-known feature of karst and while some are largely inactive or relict, being separated from local groundwater and receiving only percolating waters from above, others are active, receiving waters and sediment inputs from surface streams, including floods.

The most obvious characteristic of caves is the absence of sunlight and according to the degree of darkness and other conditions, cave environments have been divided into four zones[1], namely the:

- **twilight zone**, near the cave entrance, where light intensity, humidity and temperature vary and a large and varied fauna can be found;
- **transition zone**, with complete darkness but still variable humidity and temperature, where a number of common species live, some of which, such as bats, regularly visit the outside world;
- **deep zone**, in complete darkness and almost continual 100 % humidity and temperature, where fully cave-adapted species (troglobites) live that never venture outside the cave;
- **stagnant zone**, with complete darkness and 100 % humidity, where there is little air exchange and where carbon dioxide concentrations may become high.

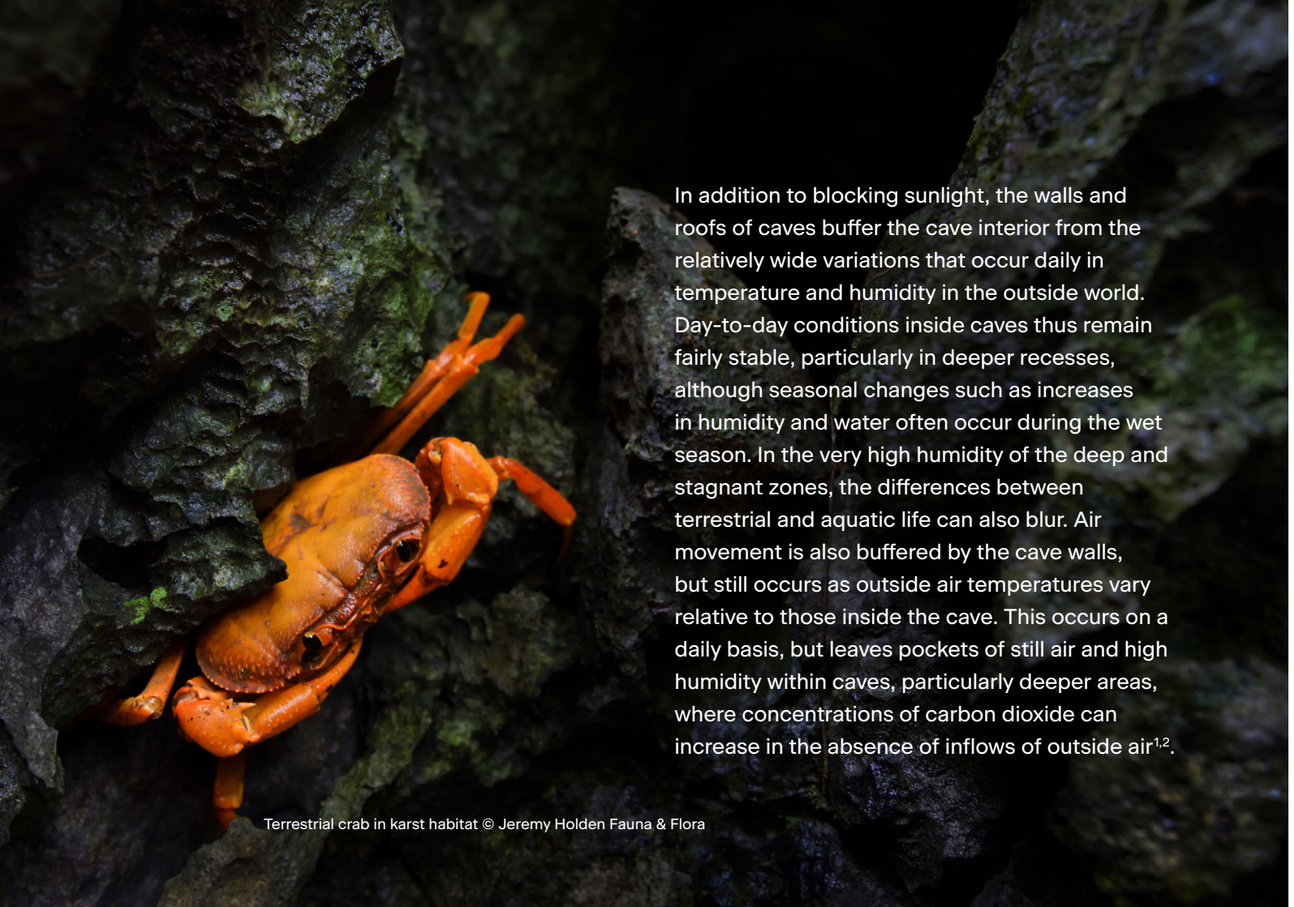
Apart from bats, which are the only mammal group to comprehensively colonize caves, the majority (>90%) of underground life consists of invertebrates. Because green plants cannot grow in the total darkness beyond the transition zone (although their roots often penetrate these areas), permanent cave dwellers are entirely dependent on materials brought in from the outside world for energy. These include organic materials falling through ceiling crevices, tree roots, guano deposited by bats and swiftlets, and drift materials washed in by streams and flood-waters[1,2]. The guano produced by bats in particular is a major source of energy in cave ecosystems and the survival of a considerable proportion of the terrestrial invertebrate fauna in tropical caves is dependent upon its continued deposition. This includes a suite of highly-adapted and narrowly-endemic arthropods (guanobionts) which complete their entire life cycles in or around bat guano[25].







▷ Cave invertebrates © Jeremy Holden Fauna & Flora

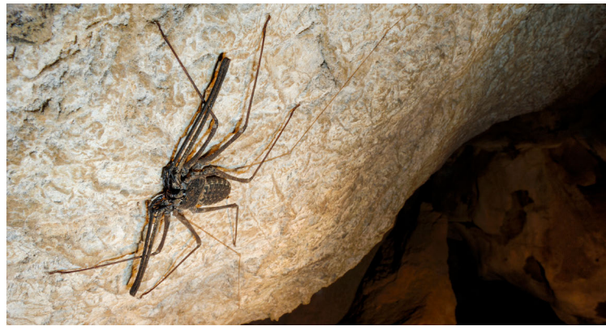


In addition to blocking sunlight, the walls and roofs of caves buffer the cave interior from the relatively wide variations that occur daily in temperature and humidity in the outside world. Day-to-day conditions inside caves thus remain fairly stable, particularly in deeper recesses, although seasonal changes such as increases in humidity and water often occur during the wet season. In the very high humidity of the deep and stagnant zones, the differences between terrestrial and aquatic life can also blur. Air movement is also buffered by the cave walls, but still occurs as outside air temperatures vary relative to those inside the cave. This occurs on a daily basis, but leaves pockets of still air and high humidity within caves, particularly deeper areas, where concentrations of carbon dioxide can increase in the absence of inflows of outside air[1,2].

Terrestrial crab in karst habitat © Jeremy Holden Fauna & Flora

Taken as a whole, the defining features of caves (total darkness, absence of plants, sporadic food availability and relatively constant climate) create one of the most unusual terrestrial ecosystems known. Cave animals have adapted in a variety of exceptional ways to cope with these extreme conditions and lost many of the features of species found aboveground. For example, their eyes are often reduced or absent (eyesight being of little use in total darkness), they have enlarged antennae or elongated appendages (to better sense the environment with), little or no pigment (so appear white or translucent), enhanced water regulation (so can expel excess water in 100% humidity without losing body salts) and have evolved longer life spans and increased fertility. These adaptations confine these species to cave interiors such that they cannot survive elsewhere.

Because karst outcrops and caves are often isolated, this has resulted in very high levels of local endemism, such that it is not uncommon for the entire population of a true cave animal (troglobite) to be confined to a single karst hill or even cave system[1,2].



Phrynichus orientalis in a Kampot cave © Jeremy Holden

Where bats are concerned, caves provide major advantages in the form of permanent, thermally -stable and humid environments which protect them against poor weather, dehydration and predators. Because roughly half of the 80+ bat species currently known in Cambodia spend at least half their lives inside caves, protection of these sites is essential for their conservation.



This poses challenges because Khmer caves are used nationwide for a wide variety of purposes[4] and any disturbance of the small and confined spaces that caves provide typically affects the entire bat colony[26]. This problem is worsened by the fact that because bats reproduce slowly (with most species producing just one offspring a year), they take a long time to recover from population losses inflicted by human activities[18]. It is also compounded by the fact that due to the abundance of Buddhist shrines in Khmer caves, the time when most bat species give birth every year is also when the number of people visiting caves annually peaks, namely during the Khmer New Year[12].

◁ *Wrinkle-lipped bats* (*Mops plicatus*) exit a cave in Battambang
© Jeremy Holden

Conservation Threats & Mitigation: First Principles

In surface environments, the size of a natural habitat is generally considered important (bigger being better), as are the extent of its links to other areas of natural habitat (habitat connectivity). In karst and subterranean habitats, isolation is positively related to levels of ende-mism, such that the most isolated limestone hills and subterranean networks tend to support the largest numbers of locally endemic species.

Since karst is highly porous, the integrity of karst caves depends on complex interactions between hydrology, biology and geomorphology within their drainage catchments. Because the health of subterranean communities is strongly influenced by their environment, activities impacting cave life include those affecting the surface environment. As a result, there is a profound need for *total catchment management* in karst landscapes to safeguard their natural processes, including holistic management of vegetation and soils[1,2].



Quarrying of limestone for cement and construction materials poses a grave threat to all karst life because it can lead to the irreversible loss of karst outcrops, leaving few options for remediation[1,18]. Demand for these materials in Southeast Asia and Cambodia has boomed in recent decades[3,27] and because very few limestone outcrops in Cambodia have been designated as protected areas to date, the future of limestone karst within the kingdom remains precarious. In Kampot for example, karst outcrops that have existed for countless millennia have been entirely destroyed by cement companies, with other limestone hills now being quarried by multiple companies and so poised to follow suit[28]. As a consequence, research is urgently needed to identify karst sites of international or national significance within the kingdom and provide an information basis for ensuring their sustainable development and management.

◁ Destruction of karst landscape in Vietnam © Jeremy Holden Fauna & Flora

Vegetation cover is crucial to the formation and maintenance of limestone karst ecosystems and survival of all karst life, including cave animals[1]. Its removal or degradation can cause severe erosion which is essentially irreversible on a human timescale and can lead to ‘rocky desertification’, a major environmental problem in China and northern Vietnam[2,29]. Allied to widespread forest loss in Cambodia, most karst outcrops within the kingdom have become isolated in agricultural landscapes and their vegetation has been widely degraded by timber extraction, fuelwood collection and fire[4,6]. This is especially true for karst hills in the Battambang, Kampot and Kep provinces and poses a conservation issue for cave bats due to their fidelity to cave roosts and because many species are confined to foraging in contiguous areas of forest[30]. As a result, retention of natural forest cover and other vegetation, both on and surrounding limestone outcrops, is essential.

▷ Balsaminaceae in karst landscape © Jeremy Holden Fauna & Flora



Cave tourism, which began in the 19th century, has dramatically increased threats to all life in caves. From around 20 million people in the mid-1990s, 150 million people worldwide visited caves for recreation in 2019 and the sector has boomed in recent decades in Southeast Asia[2,12,18]. While some cave passages can sustain considerable amounts of tourism without degradation, such as those with large exogenous rivers passing through their caverns or where sensitive areas are strictly kept off-limits to visitors, other caves have far less capacity, such as inactive or relict passages and low energy caves[1,2]. In Cambodia, cave developments for public access typically involve the installation of artificial lighting, Buddhist shrines and physical modifications to the cave substrate in the form of entrance structures, stairs, walkways and car parks[4].

Problems caused by cave development for public access include the displacement of cave-dwelling animals due to changes in light, humidity, temperature, carbon dioxide and air flow, forcing these into less optimal parts of the system; visitor trampling of less mobile cave inhabitants, such as soil-dwelling arthropods or guano-dependent species that cannot move easily; abandonment of the cave by bats due to strong light and loud noise, resulting in the loss of a major energy input and their aboveground ecosystem services; introduction of non-native species (e.g., invertebrates, fungi, algae, bacteria) to the cave by visitors that can have a severe impact on the indigenous fauna (e.g., white nose syndrome); growth of lampenflora (e.g., algae, moss, ferns and other plants) and their associated fauna due to artificial lighting; and finally, the accumulation of human garbage, which represents a new

source of food that can attract non-native biota into the cave[1,2].

The position of a cave, its passageways and their energy levels are important for assessing their vulnerability to disturbance. A distinction should be made between caves or passages that are situated above and separate from local groundwater (inactive or relict) and those which possess a continuing relationship with the water table (active). The former systems are relatively dry, relatively static in character and essentially non-renewable. Daily air movements and falling droplets of water may represent the only major events in these low energy systems and visitor impacts are likely to be considerable. Active systems that are high-energy are prone to events such as frequent flooding, causing regular modification and scouring. Speleothem development is rarer in such habitats and visitor impacts would be minimal. In moderate-energy systems, forces such as running water, persistent wind or animal activities are less active and speleothems may be abundant. Visitor impacts are more likely to become evident than in high energy systems.

When developing a cave for public access, areas that support sensitive ecosystems should be identified and maintained. This particularly applies to relict passages which have little capacity for recovery after disturbance. Until these areas are known, alterations to natural habitats in the cave should be minimised. To avoid trampling, all movement of people inside the cave should be restricted to designated pathways that avoid these locations. While speleothems may be very beautiful and should be carefully cared for (being irreplaceable on a human time scale), the most

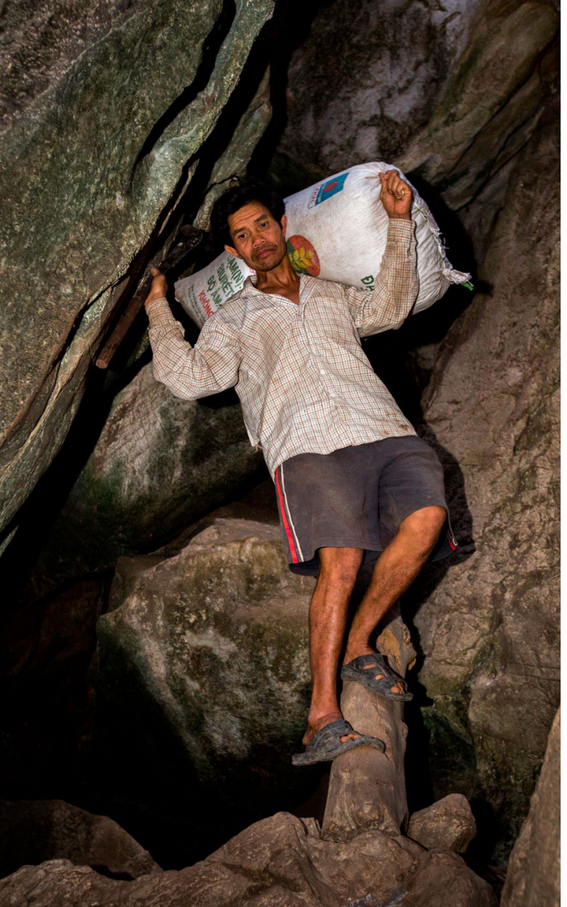


A cave in Myanmar developed without thought to ecosystem conservation © Jeremy Holden Fauna & Flora

significant part of a cave is likely to be the floor. This is because the record of a region's geo-climatic history, extinct animals and human relics is generally contained in the cave floor and may be completely unique. As such, trails are one of the most important pieces of infrastructure and should protect the cave by restricting visitors to chosen routes and preventing damage to the natural cave floor.

For similar reasons, the use of artificial lighting should be minimised within caves, with areas supporting vulnerable habitats and species left in darkness. The fundamental aim of the lighting design should be to preserve a sense of mystery and wonder, with lighting confined to low-level illumination for pathways and spot-lights on timers for specific features. Similarly, it can be beneficial to divide the cave into zones where the lights can be switched on and off by the guide, enabling only the area occupied by visitors to be lit. Additionally, it is important to use "cold" LED lights which produce less heat; other types of light encourage the growth of lampenflora which can be destructive and requires physical removal (see Online Resources for further guidance).

Despite its illegality, hunting poses a grave threat to cave bats in Cambodia because they are easily harvested in large numbers. In one nationwide survey, only ten out of 98 caves were found to be undisturbed and over half (54 caves) previously supported larger bat populations, whereas over a third (36 caves) showed evidence of hunting. As a result, the authors concluded that human activities have caused serious declines in cave bat populations throughout Cambodia.



It therefore follows that improved protection of cave biodiversity is required nationwide[4]. Other types of human disturbance often lead to a decrease in the number of bats in caves. Many studies have shown the impacts of tourism, particularly the conversion of natural cave systems to shrines[12]. Where bat colonies occur, disturbance should be minimized by preventing access to roost locations, particularly during maternity periods. At large cave colonies, access prohibitions need not extend to guano collection, which can be encouraged to support local livelihoods through sustainable harvesting techniques[31] (see Online Resources). This provides an important incentive for local communities who have effectively protected cave colonies representing the bulk of wrinkle-lipped bat populations in Cambodia for several decades[4,24].

◁ Guano harvester exiting a bat cave in Kampot © MerlinTuttle.org

Sustainable Cave Development & Management

To sustainably develop and manage a cave, the ***first*** step is to find out what is important and vulnerable by evaluating, to the fullest extent practicable, the geological, palaeontological, archaeological, cultural, biological and aesthetic aspects of the cave. This will require a thorough inventory of the cave by experienced cave specialists who should identify the features of interest, assess the vulnerability of different areas and features and identify potential uses for the cave and individual areas of the system[2]. Using this information, zones can be identified within the cave that are suitable for particular uses and these can be graded for vulnerability as follows[2]:

- **Low sensitivity.** Robust areas that can withstand all but deliberate vandalism. Suitable for all uses, provided basic precautions are employed
- **Moderate sensitivity**. Areas with interesting features that could be easily damaged unless basic precautions and care are taken, such as mandatory guides and limits on visitor numbers
- **High sensitivity**. Areas with high value, easily damaged features. Use of these areas should be minimised and measures implemented to control impacts
- **Extremely sensitive**. Areas of very high value with an elevated risk of damage. These should be strictly off-limits to visitors outside of exceptional cases such as studies that aim to improve understanding of the area or features it contains

The ***second*** step is to use this information to create a cave management plan. A good management plan recognises the significance and vulnerability of the different components of the cave and lays out a clear plan for their protection, restoration, development (infrastructure, lighting, interpretation, role of guides) and management. For caves of limited horizontal and/or vertical extent, the whole cave may be the logical management unit for many purposes. For longer caves and systems with marked variability in their values and sensitivity to impacts, a zoning approach will be more appropriate. Further guidance on cave management plans and zoning approaches are included in the 'Online Resources' section.

Once management operations have begun, the ***third*** step is to keep collecting information to create a baseline of data through visual inspection, photography, and other environmental monitoring and record-keeping activities. These should aim to reveal changes of time in the condition of resources, both impacts and improvements, and thus the effectiveness of management. Given financial limitations, monitoring efforts should prioritise features based on their importance, vulnerability and the severity of actual or potential threats. (For the same reason, these guidelines represent an ideal management scenario and should be adapted to match the available funding). To this end, it is helpful to have an inventory of the main features depicted on a cave map. Indicators chosen for monitoring need to be relevant, credible, feasible, low impact, cost effective and readily understood and undertaken by trained staff. Further guidance on cave monitoring is included in the 'Online Resources' section.

Above all, good cave management needs to:

- Recognise the position of caves within the broader context of karst and appreciate the significance and vulnerability of caves
- Maintain the natural processes, environment and cultural features of the cave and understand possible threats and ways in which damage can be prevented or reduced
- Recognise that all caves are different and that management must adapt to the needs of each cave
- Allow people to enjoy and improve their understanding of caves and provide economic and social benefits to local communities
- Actively collect information on values and threats to the cave (including visitor numbers) and use these to improve management of the cave

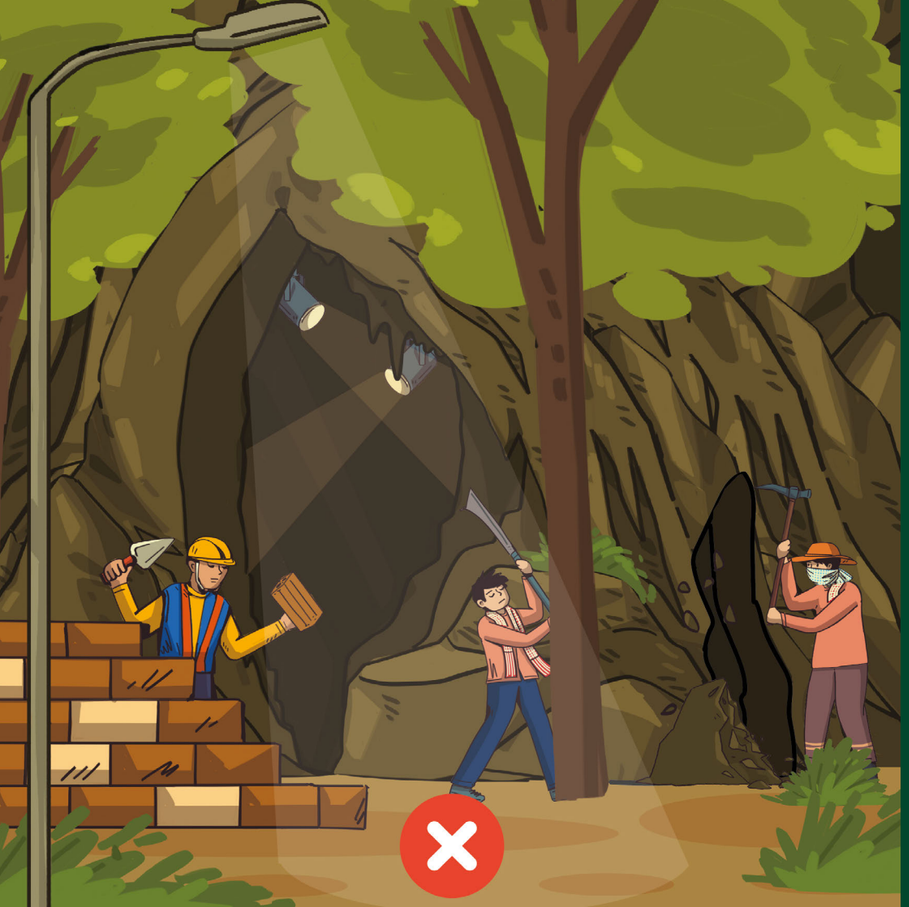


If damage to the cave environment occurs, this must be repaired and appropriate actions taken to prevent it from happening again (see Online Resources for further guidance). While administrative arrangements for cave management will vary from one area to another, effective cave management invariably requires the involvement of specialists in different fields or disciplines. For this reason, it is helpful for cave management boards to engage specialist organisations with proven experience in karst and cave conservation in an advisory capacity on an ongoing basis.

◁ Data collection in bat cave in Phnom Takriem, Battambang © Conor Wall

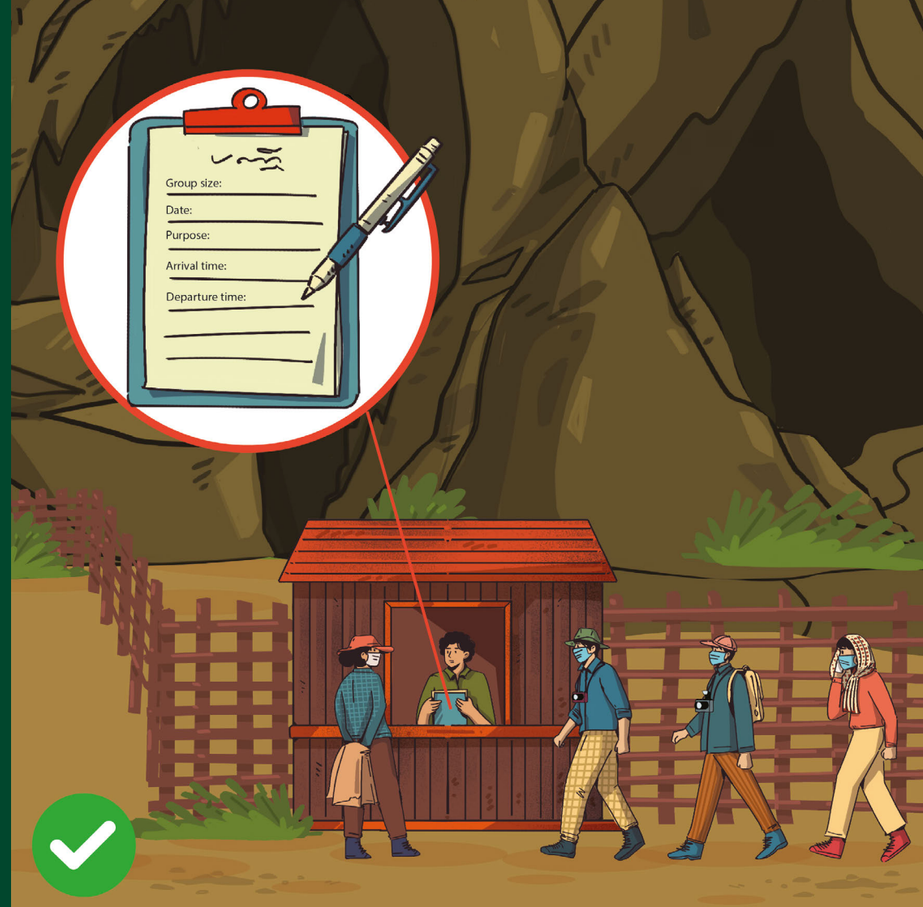
Public Caves Supporting Vulnerable Biota

Designing public access facilities for caves requires very careful thought. It must first be recognised that cave features such as speleothems and clastic sediments are essentially non-renewable resources on a human timescale. This is because the natural processes that create them occur over very long periods of time. Good cave management seeks to minimise and rectify the damage caused by public access. Consequently, determining the visitor carrying capacity of a cave requires finding a balance between providing a safe, educational and enjoyable cave experience for visitors while minimising their impacts while achieving economic goals[2].



Development and other alterations to passage morphology, speleothems, sediments and natural habitats inside a cave should be minimised, although entrance flyways for bats should be kept clear. Installation of lights and infrastructure inside the cave or nearby its entrances should also be avoided and the surrounding vegetation should be left undisturbed. It is also important not to create new entrances to the cave, because this can alter the cave microclimate and result in loss of biodiversity. The same applies to hard surfaced areas such as buildings and parking areas. These should not be positioned inside the immediate cave catchment where the natural seepage of rainwater from the surface to the cave occurs[2].

Control of public access to the cave will be required in most instances. If necessary, a fence can be installed surrounding (but never obstructing) the cave entrance. This is an appropriate place for charging entry fees and ensuring all visitors are accompanied by a guide who ensures appropriate behaviour and records the group size and the date, purpose and duration of the cave visit for management purposes. A posted fence is preferable to cave gates which some bat species do not tolerate, although fencing is more easily vandalised. If gate installation is unavoidable, it is advisable that the top section have horizontal bars with an air gap of 15 cm high and 45–75 cm wide which should allow bats to have free passage[2]. However, it is important to establish the bat-preferred design and ensure that surface waters and organic matter are not prevented from entering the cave, so as to maintain its natural energy balance.





For public access caves with bat colonies, visitors should be encouraged to wear closed-toed footwear and disposable face masks. The latter is because the most widespread health risk in caves is from histoplasmosis, an infection caused by breathing in spores of a fungus often found in bird and bat droppings. Visitors should also be encouraged to clean and disinfect their footwear before entering and on exiting the cave. This is to prevent the introduction of non-native species that can have a severe impact on the indigenous cave fauna. For example, millions of cave bats in the USA have died due to an infectious fungal disease (white nose syndrome) which is believed to have been introduced to caves there by an overseas visitor with contaminated footwear. Like other mammals, bats can carry rabies and transmit this to humans through a bite or scratch by an infected animal. While over 99% of human cases are caused by dogs, cave visitors should avoid handling bats and if accidentally bitten, promptly disinfect the wound and seek medical advice on post-exposure prophylaxis.

Cave visitors should be allowed to see and appreciate, but they should not be allowed to touch. Since prevention is always better than a cure, it is best to keep visitors away from delicate cave features. These are very easily damaged and non-renewable on a human timescale. Tourist routes within the cave should always be designed to have a minimum impact on cave habitats and features. Similarly, visitor group sizes should be kept small. Noise should be kept to a minimum (e.g., no machinery, music players, loud speakers, shouting, etc.). This is particularly important if bats are present in the cave, because they are highly sensitive to non-tactile human disturbance[26]. Measures should also be taken to remove and prevent the disposal of human garbage inside the cave because this represents a new source of food that can attract non-native species into the cave.

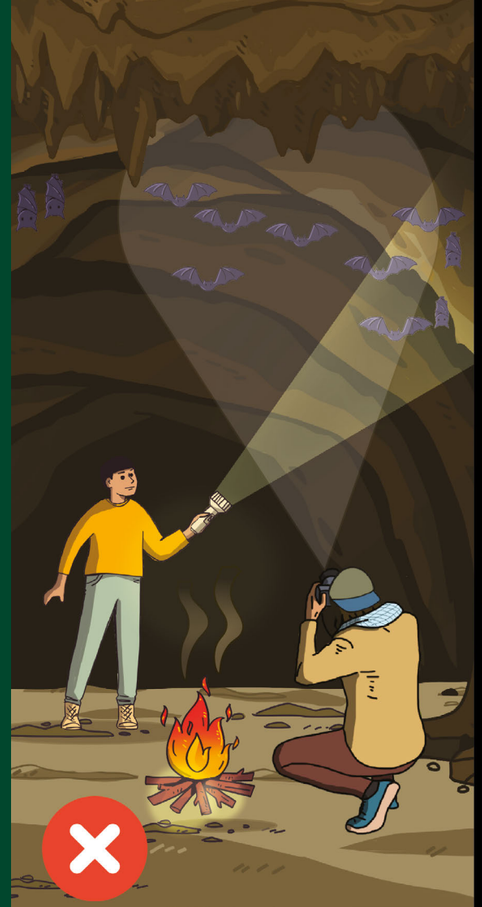






Disturbance during pregnancy, lactation and weaning is highly detrimental to bat populations. As a result, visitors should not be allowed to access bat roosts during maternity periods when they are exceptionally vulnerable to human disturbance. March–July is the primary reproductive period for most bats in Cambodia, although there are several cave-dwelling species that have two birth periods a year. These species tend to form the largest colonies. For example, the critical period for recruitment of *Taphozous* spp. and *Mops plicatus* in Cambodia is March–July and September–December[12,24]. Outside of breeding seasons, guides should ensure that all visitors remain at least 3m and preferably much more from roosting bats. This is because research suggests that maintaining a distance of at least 12m between tourists and bats causes the least disturbance[32].

For similar reasons, visitors should not be allowed to light fires, incense or allow the use of lamps that produce fumes or smoke inside a cave or nearby the cave entrances. In concentrating in highest portions of the cave and being generally toxic or simply depleting oxygen, these are detrimental to bats which tend to roost in the same locations. Because research has shown that avoiding illumination of bats causes the least disturbance[32], cave visitors should not be allowed to shine lights directly on the bats or to use spotlights, flash photography or airborne drones when bats are emerging from the cave entrances in the evening. Because the flight paths of bats are erratic and cannot be predicted with any certainty, the latter poses an obvious risk of collisions and could otherwise disrupt the evening emergence through engine noise alone.







Wrinkle-lipped bat (*Mops plicatus*) © Charles Francis

Cave Bat Colonies Harvested for Guano

Harvesting of cave bat guano for plant fertilizer is common in Cambodia, where its sale and use features prominently in many local economies[4,18]. Although this activity can destroy palaeo-environmental archives and damage the invertebrate communities that depend on guano[2], it is recognised that continued harvesting is inevitable as it provides an important source of fertilizer and income in rural areas. As such, these guidelines seek to promote practices that minimize harvesting impacts to the guano producers (bats) and the cave systems and species that depend on its regular deposition[31]. Most if not all of the actions recommended for caves open to the public will apply to caves harvested for bat guano and further guidance is available in the 'Online Resources' section, particularly IUCN SSC (2014)[31].

Human disturbance represents one of the greatest threats to cave bats worldwide and any human presence inside the relatively small and confined spaces that caves provide typically disturbs the entire bat colony. Because nationwide surveys in Cambodia have shown that around 90% of caves within the country are used for a variety of purposes by humans, including over a third for guano harvesting, human disturbance has led to widespread declines in cave bats nationally and represents a serious conservation issue throughout the kingdom[4]. Due to the major contribution guanobionts make to overall cave diversity, disturbance to bats is also increasingly regarded as one of the most serious threats to tropical cave invertebrates[25]. Consequently, guano harvesting operations should attempt to minimize human disturbance by undertaking these at night whenever possible after most or all of the bats have already left the cave to forage.





As mentioned, human disturbance during pregnancy, lactation and weaning is highly detrimental to recruitment in bat populations. Even non-tactile disturbance during these periods can lead to: 1) outright death of young that lose their roost-hold and fall to the cave floor, 2) females abandoning the roost for less ideal sites where prospects for reproductive success may be reduced, 3) greater energy expenditure among females and less efficient energy transfer to young (translating into slower growth of young and increased foraging demands on females)[26]. Guano collection should therefore be avoided whenever possible during maternity periods when young bats and their mothers are present in the cave (or only harvested at night after the bats have already left, as above) because these are especially weak and vulnerable to human disturbance[24].

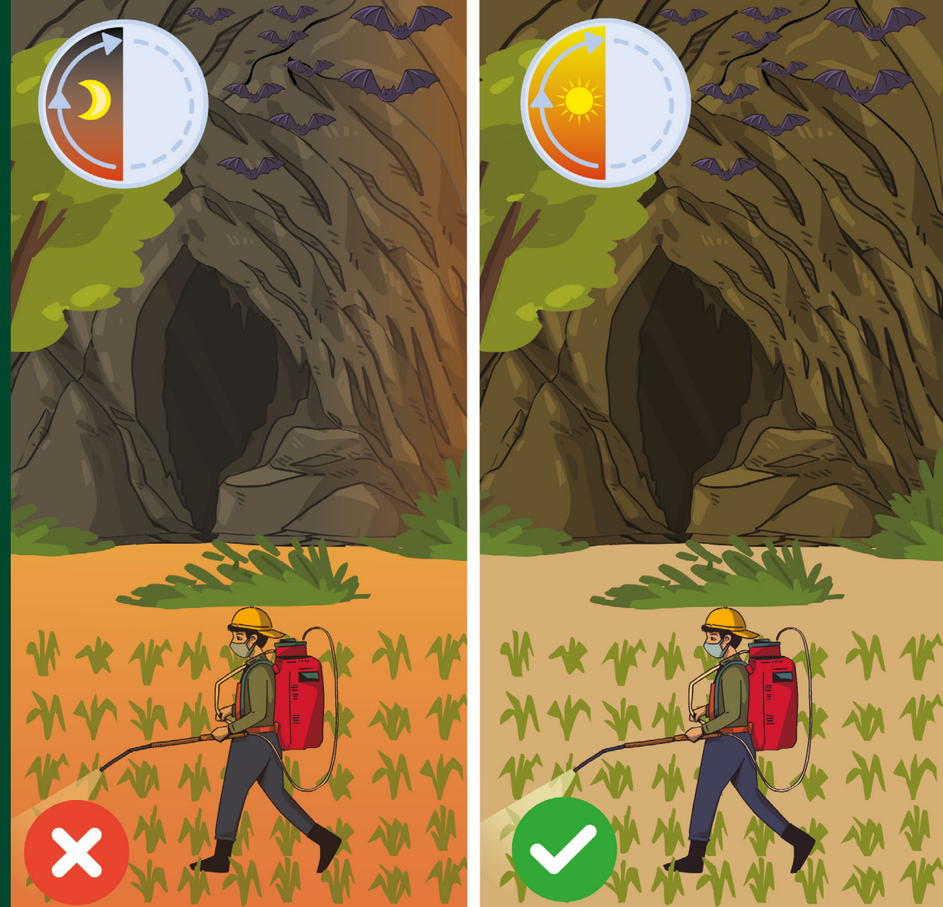
To reduce disturbance to less mobile cave inhabitants, such as soil-dwelling arthropods or guano-dependent species that cannot move easily, all movement and traffic of people inside the cave should be restricted to designated routes and the location of guano collection points should be rotated regularly. This will help reduce impacts to the highly-specialized and often endemic invertebrates that depend heavily on guano for their survival and minimize trampling of these habitats. At least 10% of the guano present should be left undisturbed in place during each harvesting session to provide some measure of continuity for these animals.





In the interests of their health, all harvesters should wear dedicated clothing, closed-toed footwear and disposable gloves and masks when collecting and handling guano. Eye protection is also recommended. To ensure good hygiene and reduce risks of cross-cave contamination, clothes and footwear should be thoroughly washed after each harvesting session and as before, wearing of face masks is advised because histoplasmosis may pose the greatest risk in poorly-ventilated caves containing large guano deposits. Like other mammals, bats can carry rabies and transmit this to humans through a bite or scratch by an infected animal. While >99% of human cases are caused by dogs, harvesters should avoid handling bats and if accidentally bitten or otherwise injured, promptly disinfect the wound and seek medical advice on post-exposure prophylaxis.

Bat populations are vulnerable to agricultural pesticides, which they often consume in pest insects that are targeted by farmers. These toxins accumulate in bats, causing lethal and sublethal impacts. Organochlorine, organophosphate and carbamate insecticides can pose serious threats to bats. Studies suggest that some synthetic pyrethroids (e.g., permethrin) do not result in obvious harm, but other forms may[31]. As a result, spraying of pesticides at times and places where bats are likely to be harmed should be minimized — especially in areas where bats habitually forage. Spraying of crops at dawn should be favoured because this is less likely to contaminate insects that are eaten by bats.





Since the rate of guano accumulation and related harvesting records can reflect the size of a bat colony in a given cave, these can be included in monitoring efforts for the bat population — provided the data are appropriately collected and interpreted, and their limitations clearly understood. Guano harvesting operations are advised to keep written records for each harvest, including the following data:

- Date of each harvest
- Harvest quantity (number of rice sacks or other measure)
- Number of harvesters
- Harvest locations
- Harvester remarks regarding the numbers of bats, signs of disturbance and other observations

Online Resources

- Guidelines for Cave and Karst Protection (2nd edition): https://engage.iucn.org/system/files/2023-04/Guidelines-for-Cave-and-Karst-Protection-2022.pdf
- Biodiversity and Cultural Property in the Management of Limestone Resources: Lesson from East Asia: https://documents1.worldbank.org/curated/en/804721468772149189/pdf/multi-page.pdf
- Living with Karst: a Fragile Foundation: https://www.americangeosciences.org/static/files/profession/EAS_Living_With_Karst_A_Fragile_Foundation.pdf
- Encyclopaedia of Caves and Karst Science: https://sudartomas.wordpress.com/wp-content/uploads/2012/11/encyclopediaof_cavesandkarstscience.pdf
- International Guidelines for the Development and Management of Show Caves: https://uis-speleo.org/wp-content/uploads/2020/03/Guidelines-for-Show-Caves-ISCA-IUCN-UIS.pdf
- Cave Conservation and Restoration: https://caves.org/wp-content/uploads/2022/06/CCR-Book.pdf
- IUCN SSC Guidelines for Minimizing the Negative Impact to Bats and Other Cave Organisms from Guano Harvesting: https://portals.iucn.org/library/efiles/documents/Rep-2014-002.pdf
- IUCN SSC Cave Invertebrate Specialist Group: https://caveinvertebrates.org/
- Global Union of Bat Diversity Networks: https://www.gbatnet.org/
- IUCN SSC Bat Specialist Group: https://www.iucnbsg.org
- International Union of Speleology: https://uis-speleo.org/

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These guidelines were developed in response to a growing need for practical, tools to inform the conservation and sustainable development of Cambodian caves. They are intended to support local authorities, protected area managers, and other decision-makers to balance the use and protection of caves - ensuring that communities can benefit from them today without compromising their future.